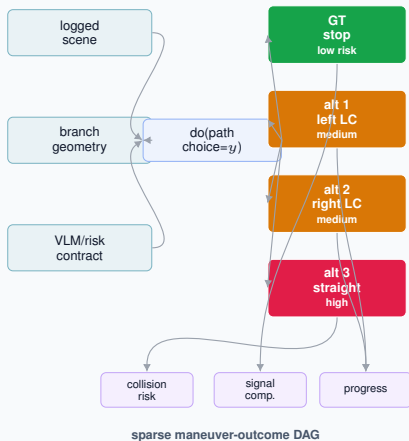


CounterDrive Framework

geometry-grounded alternatives + causal contract + semantic-control Adv-BMT generator

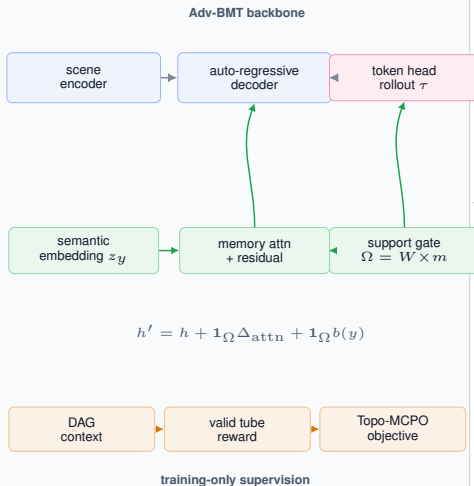
1. Expand local alternatives

one logged decision becomes factual + semantic branches



2. Generate with semantic-control MotionLM

pretrained Adv-BMT traffic prior plus lightweight adapters



3. Score rollouts and train agents

generated counterfactuals become downstream data

